

ORLANDO MELBOURNE INTL, FL, USA

WMO: 722040

Lat: 28.101N

Lon: 80.644W

Elev: 27

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 12838

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	38.7	43.1	25.0	19.1	46.6	30.0	24.2	49.3	24.8	63.7	21.7	62.5	7.8	310	0.402

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.6	91.8	77.6	90.5	77.8	89.6	77.9	80.4	87.7	79.8	87.2	79.1	86.4	10.5	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.7	149.0	84.1	77.4	142.8	83.7	76.9	140.4	83.4	43.7	87.5	43.0	86.8	42.5	86.7	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.8	18.8	17.4	DB	32.8	96.0	3.7	1.9	30.1	97.4	28.0	98.5	26.0	99.6	23.3	101.0
			WB	29.5	82.6	3.6	1.8	26.9	83.9	24.7	85.0	22.7	86.0	20.1	87.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	73.5	61.8	64.6	68.0	72.4	77.2	80.8	82.0	82.3	81.1	76.6	69.5	65.3
	DBStd	9.02	8.22	7.89	6.77	5.10	3.41	2.49	2.13	2.08	2.33	5.04	6.40	8.07
	HDD50	21	12	4	0	0	0	0	0	0	0	0	0	5
	HDD65	452	158	95	52	9	0	0	0	0	0	5	33	100
	CDD50	8601	378	414	557	671	843	923	993	1000	932	824	586	480
	CDD65	3555	59	85	143	231	378	473	528	535	482	363	168	110
	CDH74	31072	241	402	734	1517	3175	4578	5545	5637	4828	3078	894	443
	CDH80	9820	21	56	130	290	852	1627	2199	2187	1632	687	103	36
(14) Wind	WSAvg	8.6	8.6	8.8	9.7	9.8	9.5	7.9	7.1	7.3	8.1	9.4	8.9	8.3
(15) Precipitation	PrecAvg	51.9	2.3	2.0	3.0	2.3	3.3	6.7	6.2	7.4	7.1	5.3	2.3	2.4
	PrecMax	69.7	6.1	6.6	11.7	5.8	10.1	14.3	11.7	24.0	14.2	11.4	8.1	9.0
	PrecMin	37.9	0.0	0.0	0.0	0.6	0.7	0.1	2.2	1.5	3.5	0.7	0.2	0.4
	PrecStd	9.6	1.4	1.4	2.9	1.3	2.3	3.2	2.6	4.3	2.9	3.1	1.7	2.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.3	85.0	87.9	89.5	92.1	94.5	93.4	93.2	91.9	89.1	85.5	83.2
		MCWB	70.4	69.9	71.1	71.7	74.1	77.6	78.4	78.3	77.4	77.1	74.5	72.7
	2%	DB	79.4	81.5	83.7	85.6	89.1	91.2	91.3	91.0	89.7	86.9	82.5	80.9
		MCWB	69.3	70.0	70.1	72.5	73.8	77.4	78.2	78.8	77.9	76.4	73.2	71.9
	5%	DB	76.7	79.0	80.9	82.9	86.3	89.5	90.1	90.0	88.2	85.0	81.0	78.9
		MCWB	68.0	69.1	69.4	71.3	74.0	77.6	78.3	78.8	77.5	75.5	72.1	70.8
	10%	DB	73.8	75.9	78.6	81.2	84.2	87.7	88.5	88.4	87.1	83.6	79.1	76.4
		MCWB	66.9	67.7	68.1	70.2	73.5	76.9	78.0	78.4	77.2	74.6	70.3	69.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.2	73.4	74.8	76.1	77.6	80.4	81.1	81.4	80.8	79.7	77.0	75.1
		MCDB	79.0	79.1	81.0	83.5	85.8	89.3	88.9	88.3	87.1	85.0	81.9	80.3
	2%	WB	71.5	72.0	73.0	74.6	76.5	79.2	80.1	80.4	79.8	78.2	75.4	73.6
		MCDB	75.9	78.0	79.0	81.5	84.3	87.8	88.2	87.6	86.2	83.7	80.4	78.2
	5%	WB	69.9	70.6	71.4	73.4	75.6	78.3	79.3	79.8	78.7	77.4	74.0	72.6
		MCDB	74.6	76.6	77.9	79.9	83.1	86.4	87.6	87.1	85.1	82.9	79.0	76.9
	10%	WB	68.4	69.3	70.0	72.2	74.9	77.7	78.5	79.1	78.1	76.4	72.6	71.1
		MCDB	73.0	75.0	76.3	78.9	82.1	85.3	86.3	86.2	84.4	81.9	77.4	75.2
(35) Mean Daily Temperature Range	5% DB	MDBR	18.9	18.2	17.8	16.6	15.0	14.1	14.2	13.6	12.5	13.1	15.9	16.7
		MCDBR	18.2	18.0	18.5	17.8	18.2	17.0	15.7	14.8	13.7	13.0	14.6	15.5
		MCWBR	10.1	8.8	8.6	7.9	7.4	6.8	6.4	6.0	5.5	5.6	7.5	8.5
	5% WB	MCDBR	16.1	16.1	16.4	15.4	15.4	14.8	14.4	13.6	12.8	11.4	13.0	13.8
		MCWBR	10.1	8.8	8.4	7.8	6.9	6.7	6.5	6.2	5.7	5.4	7.4	8.4
(40) Clear-Sky Solar Irradiance	taub		0.346	0.359	0.373	0.396	0.427	0.454	0.472	0.461	0.426	0.401	0.364	0.353
	taud		2.532	2.509	2.458	2.373	2.328	2.328	2.276	2.325	2.415	2.452	2.504	2.535
	Ebn at Noon		285	290	291	286	275	266	261	264	271	272	276	277
	Edh at Noon		29	31	35	39	41	41	43	41	36	33	29	27
(44) All-Sky Solar Radiation	RadAvg		1079	1311	1634	1929	1984	1863	1858	1774	1559	1390	1136	980
	RadStd		77	76	104	75	141	169	104	107	91	89	59	67

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.70	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+328	+284
(47) Regional (31 neighbors)	+0.77	N/A	N/A	N/A	N/A	N/A	N/A	-74	+283	+205

Nomenclature: See separate page